

ISC Assignment-03

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Question 1: Write a python program to perform Caesar Cipher brute force decryption by showing all possible 25 decryptions.

Code:

```
# PART 1: BRUTE FORCE CAESAR CIPHER
```

```
def caesar_decrypt(text, key):
```

```
    result = ""
```

```
    for ch in text:
```

```
        if ch.isupper():
```

```
            result += chr((ord(ch) - ord('A') - key) % 26 + ord('A'))
```

```
        elif ch.islower():
```

```
            result += chr((ord(ch) - ord('a') - key) % 26 + ord('a'))
```

```
        else:
```

```
            result += ch
```

```
    return result
```

```
def brute_force_caesar(ciphertext):  
    print("All possible decryptions:\n")  
  
    for key in range(1, 26):  
        print("Key", key, ":", caesar_decrypt(ciphertext, key))  
  
    print("==== CAESAR CIPHER ====")  
  
    ciphertext = input("Enter Caesar ciphertext: ")  
  
    brute_force_caesar(ciphertext)
```

Output:-

```
• PS D:\PROGRAMMING\programms\ISC> cd "d:\PROGRAMMING\programms\ISC\Lab-3" ; python Q1.py
===== CAESAR CIPHER =====
Enter Caesar ciphertext: KLWHVK
All possible decryptions:

Key 1 : JKVGUJ
Key 2 : IJUFTI
Key 3 : HITESH
Key 4 : GHSDRG
Key 5 : FGRCQF
Key 6 : EFQBPE
Key 7 : DEPAOD
Key 8 : CDOZNC
Key 9 : BCNYMB
Key 10 : ABMXLA
Key 11 : ZALWKZ
Key 12 : YZKVJY
Key 13 : XYJUIX
Key 14 : WXITHW
Key 15 : VWHSGV
Key 16 : UVGRFU
Key 17 : TUFQET
Key 18 : STEPDS
Key 19 : RSDOCR
Key 20 : QRCNBQ
Key 21 : PQBMAP
Key 22 : OPALZO
Key 23 : NOZKYN
Key 24 : MNYJXM
Key 25 : LMXIWL
```

Question 2: Write a python program to encrypt and decrypt patient diagnostic notes using a Monoalphabetic Substitution Cipher.

Code:

```
# PART 2: MONOALPHABETIC SUBSTITUTION CIPHER
```

```
def mono_encrypt(text, key):  
    result = ""  
  
    for ch in text:  
        if ch.isupper():  
            result += key[ord(ch) - ord('A')]  
        elif ch.islower():  
            result += key[ord(ch) - ord('a')].lower()  
        else:  
            result += ch  
  
    return result
```

```
def mono_decrypt(text, key):
```

```
result = ""
```

```
rev = {}
```

```
for i in range(26):
```

```
    rev[key[i]] = chr(ord('A') + i)
```

```
for ch in text:
```

```
    if ch.isupper():
```

```
        result += rev[ch]
```

```
    elif ch.islower():
```

```
        result += rev[ch.upper()].lower()
```

```
    else:
```

```
        result += ch
```

```
return result
```

```
print("==== MONOALPHABETIC SUBSTITUTION CIPHER =====")
```

```
key = "QWERTYUIOPASDFGHJKLZXCVBNM"
```

```
print("Substitution Key:", key)
```

```
text = input("Enter patient data / diagnostic note: ")
```

```
encrypted = mono_encrypt(text, key)
```

```
decrypted = mono_decrypt(encrypted, key)
```

```
print("\nOriginal :", text)
```

```
print("Encrypted:", encrypted)
```

```
print("Decrypted:", decrypted)
```

Output:-

```
• PS D:\PROGRAMMING\programms\ISC> cd "d:\PROGRAMMING\programms\ISC\Lab-3" ; python Q2.py
===== MONOALPHABETIC SUBSTITUTION CIPHER =====
Substitution Key: QWERTYUIOPASDFGHJKLZXCVBNM
Enter patient data / diagnostic note: Patient Hitesh: High blood pressure, heart rate 72 bpm.
Original : Patient Hitesh: High blood pressure, heart rate 72 bpm.
Encrypted: Hqzotfz Ioztli: Ioui wsggr hktllxkt, itqkz kqzt 72 whd.
Decrypted: Patient Hitesh: High blood pressure, heart rate 72 bpm.
```